- They are a group of glycoproteins present in the serum and tissue fluids of all mammals.
- They are secreted by activated B cells and plasma cells in response to exposure to an antigen (immunogen).
- They recognize and bind to a particular epitope with very high specificity.
- Immunoglobulins are either membrane-bound or soluble in serum

تم المسح الضوئي بـ CamScanner

STRUCTURE OF ANTIBODY

- The simplest antibodies are "Y"-shaped macromolecules composed of 4 chains: 2heavy (H) chains &2 light (L) chains

- Each H and L chains consists of a variable region and a constant region.

- The constant regions of the same immunoglobulin class have the same amino acid sequence and are responsible for the characteristic functions and properties of the different classes of immunological molecules.

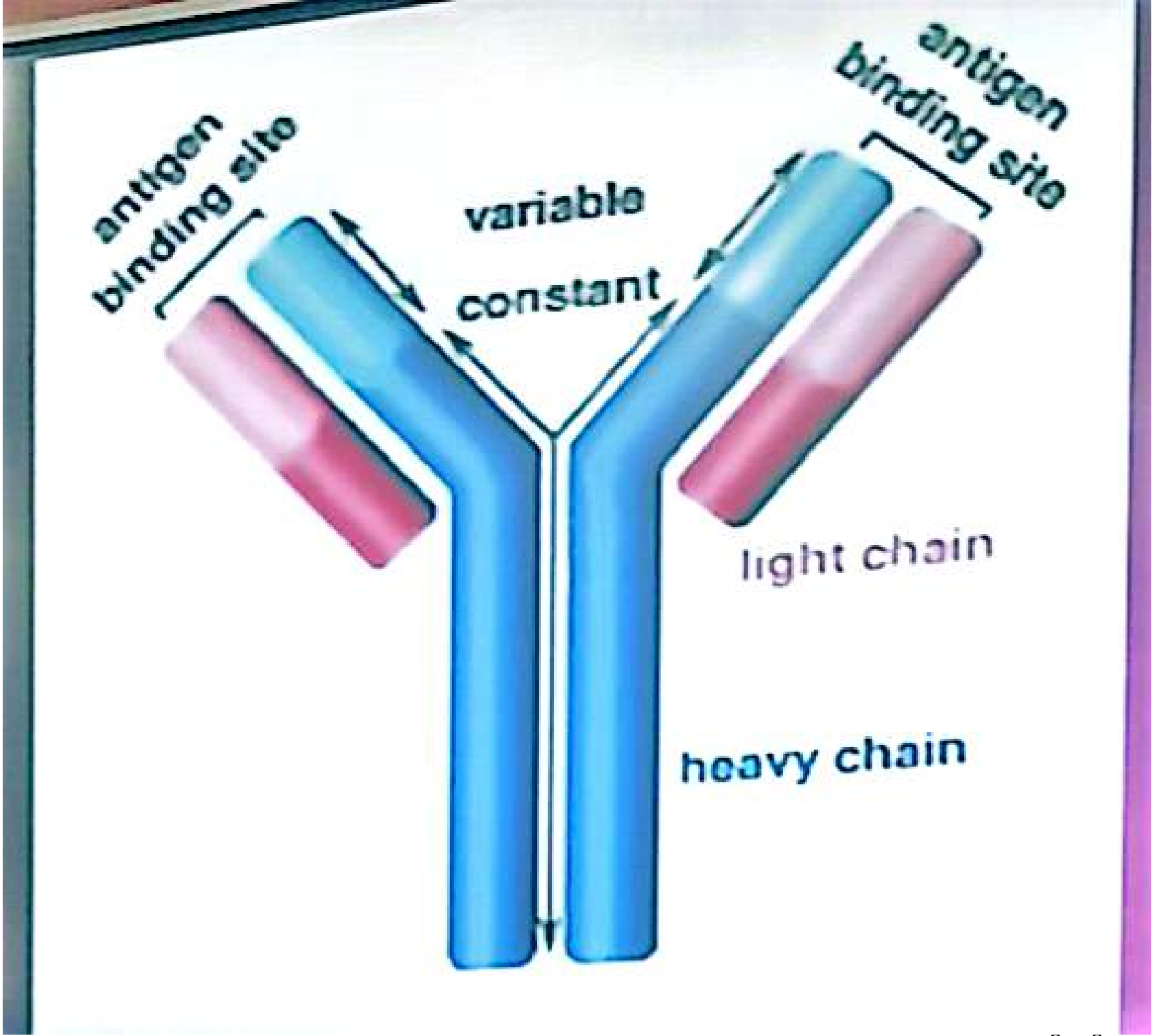
تم المسح الضوئي بـ CamScanner

- The constant region (Fc) determines the class of Ig (IgM, IgG, IgA, IgD, and IgE).

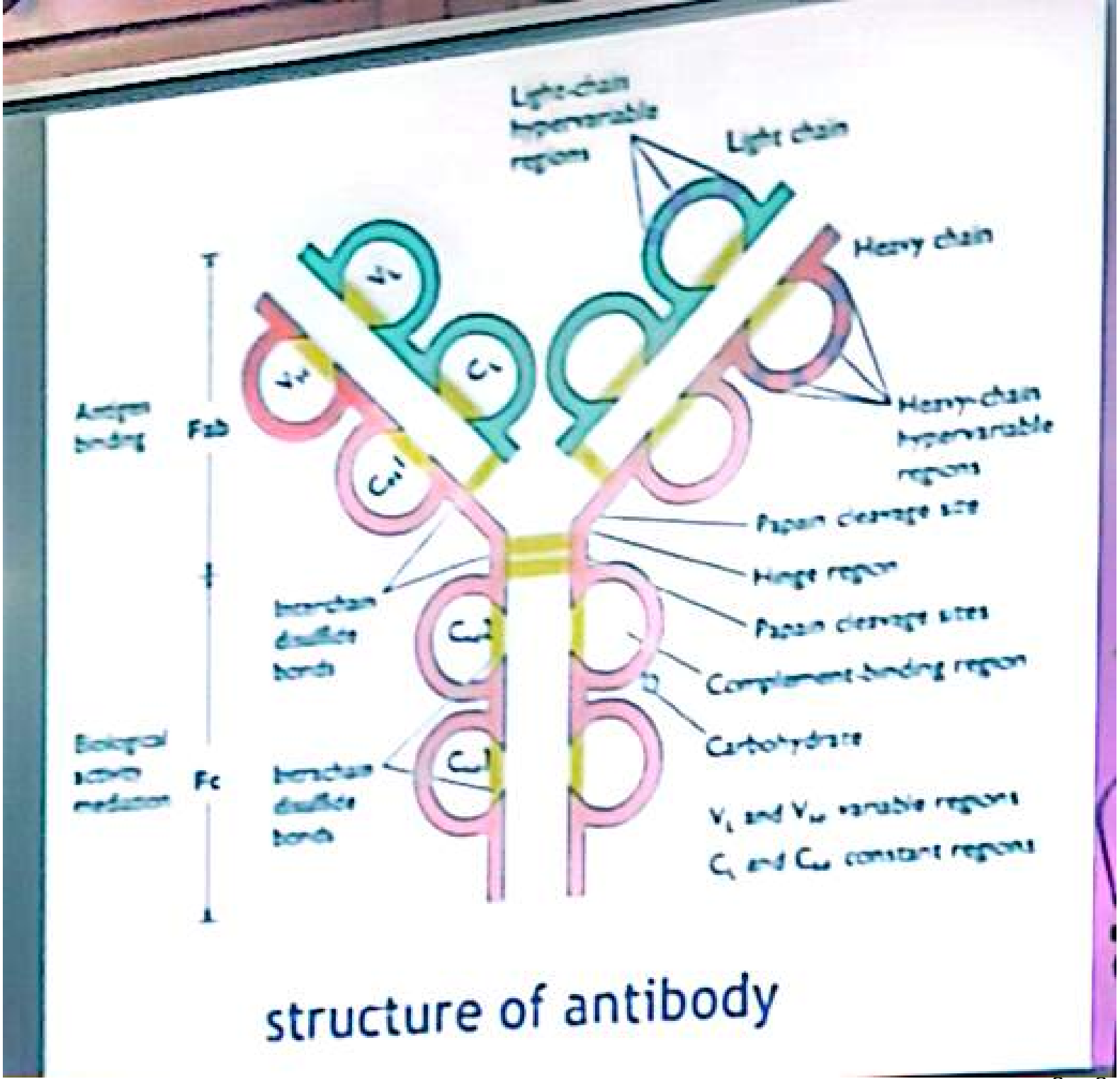
- On the other hand, the variable region interacts with the antigenic determinants on antigens so it is called Ag binding region(Fab).

- Since each antigenic determinant is different from another, the variable regions of the antibody molecules are also different in their arrangement of a.a. sequences

تم المسح الضوئي بـ CamScanner



تم المسح الضوئي بـ CamScanner



structure of antibody

تم المسح الضوئي بـ CamScanner

CLASSES OF IMMUNOGLOBULINS

- There 5 classes of antibodies. They are differ in structure and biological activity. Each class of antibody is called isotype of Ab.

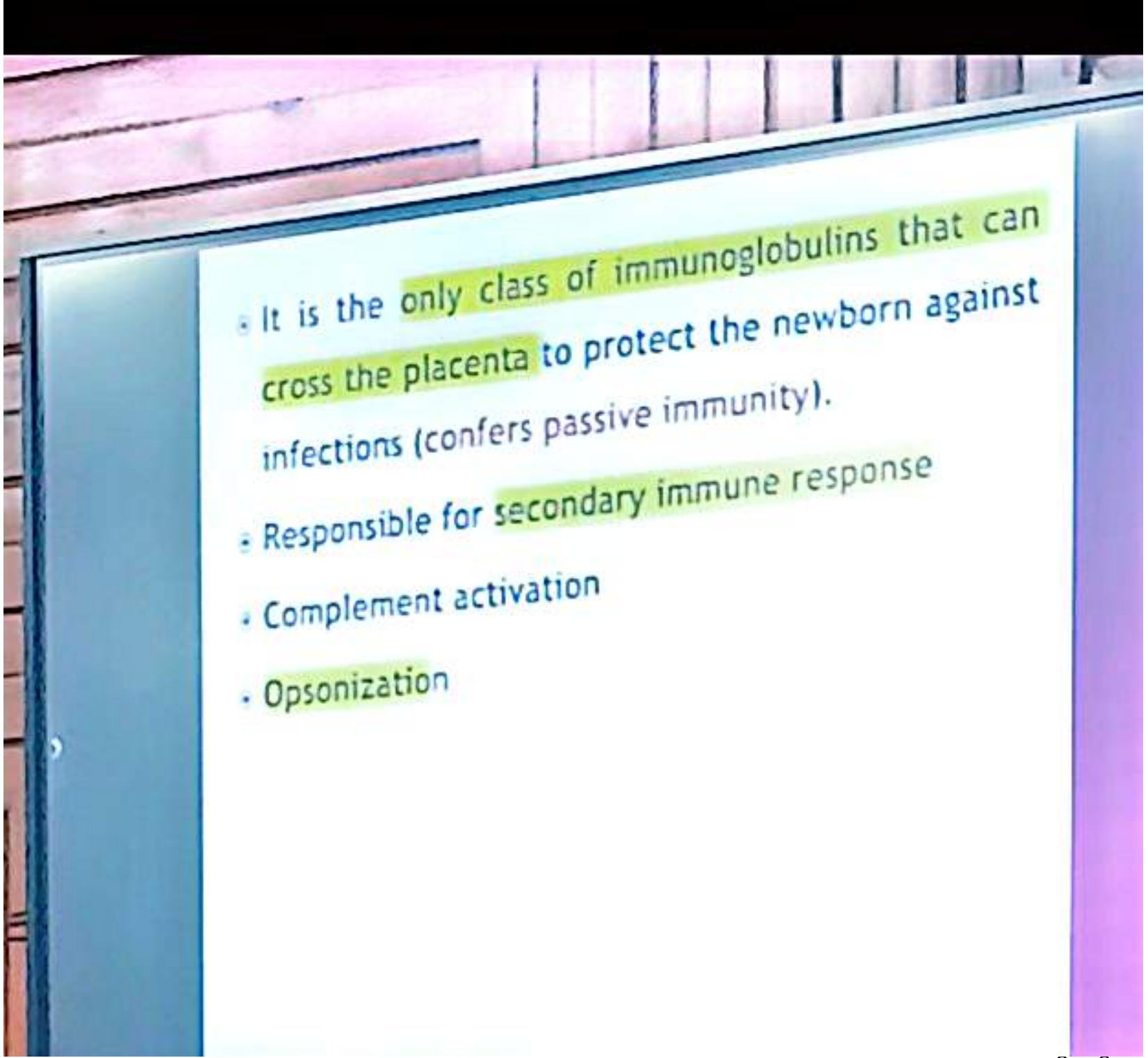
1) IgD:

It is monomer attached to surface of B cells. It functions during B cell development, maturation and activation.

2) IgG: (4 subclasses, IgG1-4)

- It is monomer and has 2 epitope-binding sites. It is the major circulating antibody found in the blood, lymph, and tissue fluids, and comprises approximately 80% of the serum immunoglobulins after age of two years.

تم المسح الضوئي بـ CamScanner



تم المسح الضوئي بـ CamScanner

- Which of the following immunoglobulins can cross placenta?

- a. IgG
- b. IgM
- c. IgD
- d. IgE
- e. IgA

تم المسح الضوئي بـ CamScanner

3) IgM:

- It is pentamer (5 unites linked by J-chain) released by plasma cells during the primary immune response.
- It is able to interact with complement but is unable to cross the placenta.
- It is the first immunoglobulin produced by human fetus.
- Powerful agglutinble
- Complement activation
- B cell receptor

تم المسح الضوئي بـ CamScanner

. Which of the following immunoglobulins present in high concentration in mucous secretion ?

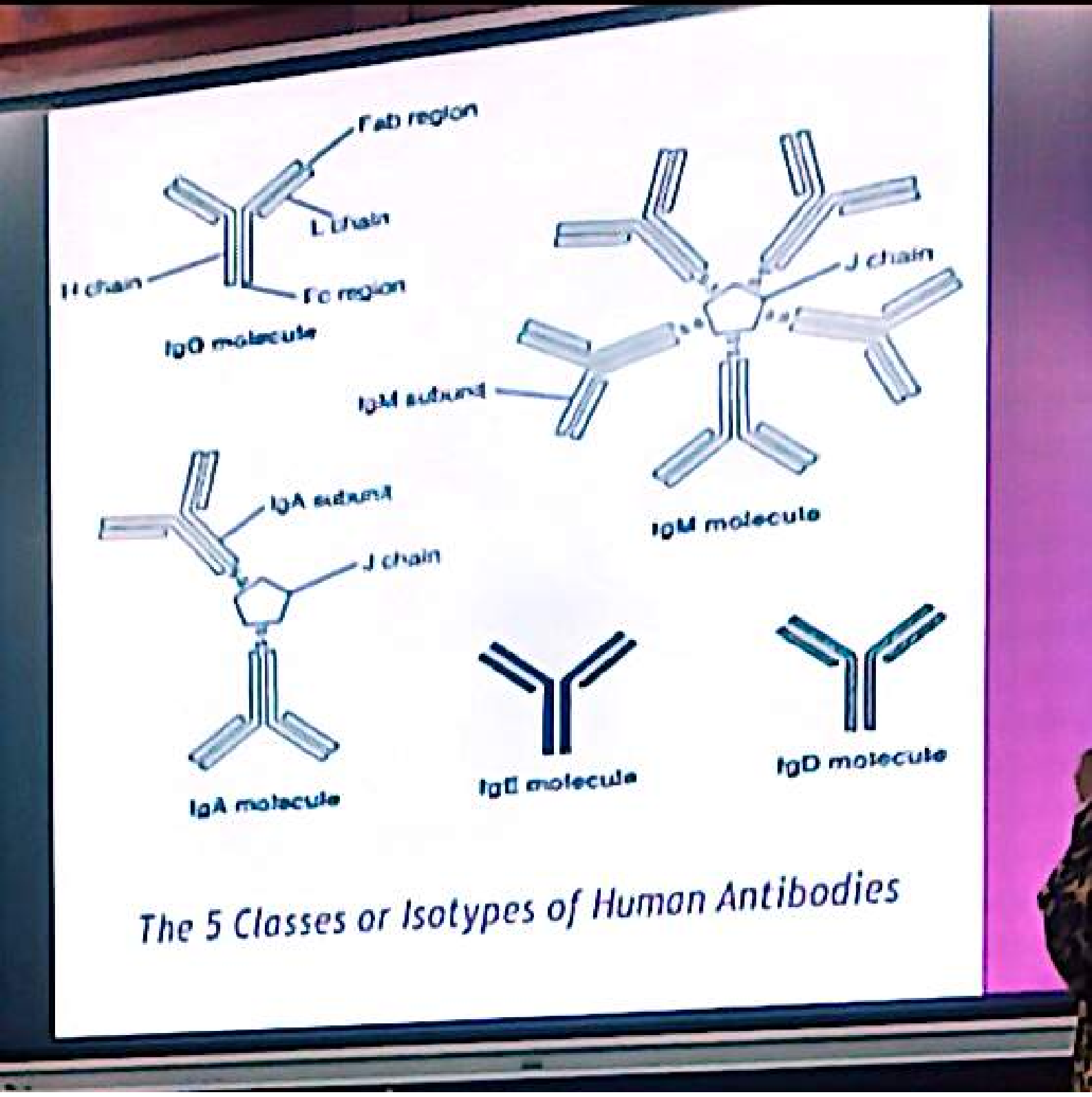
- a. IgG
- b. IgM
- c. IgD
- d. IgE
- e. IgA

تم المسح الضوئي بـ CamScanner

5) IgE:

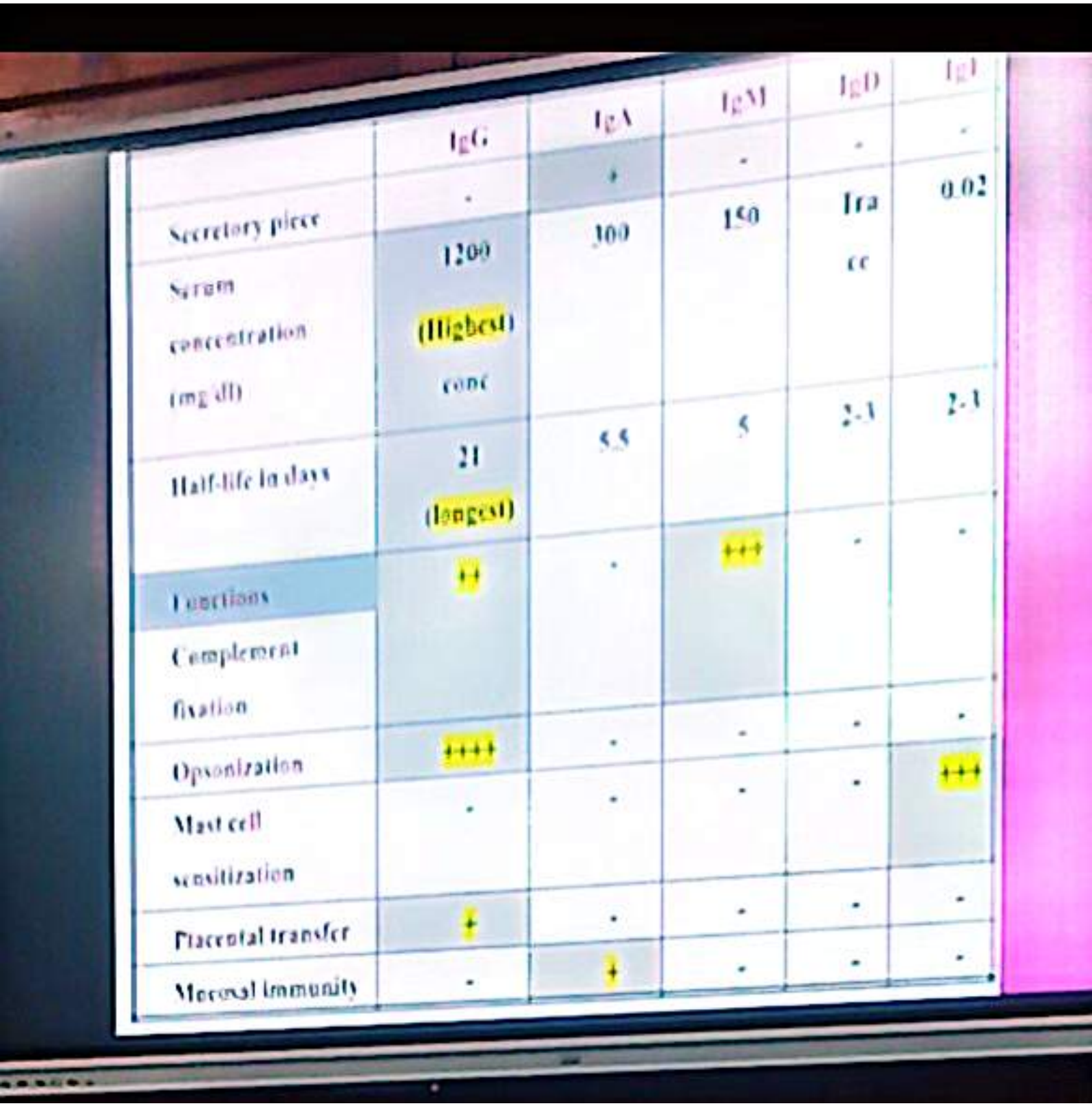
- It is monomer, It is present in trace amounts in serum.
- It has a half-life of 2 days in serum but the membrane bound form remains longer
- It is responsible for many allergic and hypersensitivity reactions because it binds to mast cells and basophils, causing histamine release when activated.
- Its serum level increase in allergic subjects and in parasitic infection.

تم المسح الضوئي بـ CamScanner



The 5 Classes or Isotypes of Human Antibodies

تم المسح الضوئي بـ CamScanner



	IgG	IgA	IgM	IgD	IgE
Secretory piece	-	+	-	-	-
Serum concentration (mg/dl)	1200 (Highest) conc	300	150	Tra ce	0.02
Half-life in days	21 (longest)	5.5	5	2-3	2-3
Functions					
Complement fixation	++	-	+++	-	-
Opsonization	++++	-	-	-	-
Mast cell sensitization	-	-	-	-	+++
Placental transfer	+	-	-	-	-
Mucosal immunity	-	+	-	-	-

تم المسح الضوئي بـ CamScanner

FUNCTIONS OF IMMUNOGLOBULINS(IMP)

1-Opsonization:

Antibodies facilitate phagocytosis

2. Neutralization:

Antibodies bind to and block specific sites on viruses or exotoxins, thus preventing these antigens from binding to receptors on tissue cells.

3. Agglutination:

Antibodies bind the same determinant on more than one antigen on cell membrane. Cell-bound antigens are cross-linked clumping (agglutination)

تم المسح الضوئي بـ CamScanner

4. Precipitation:

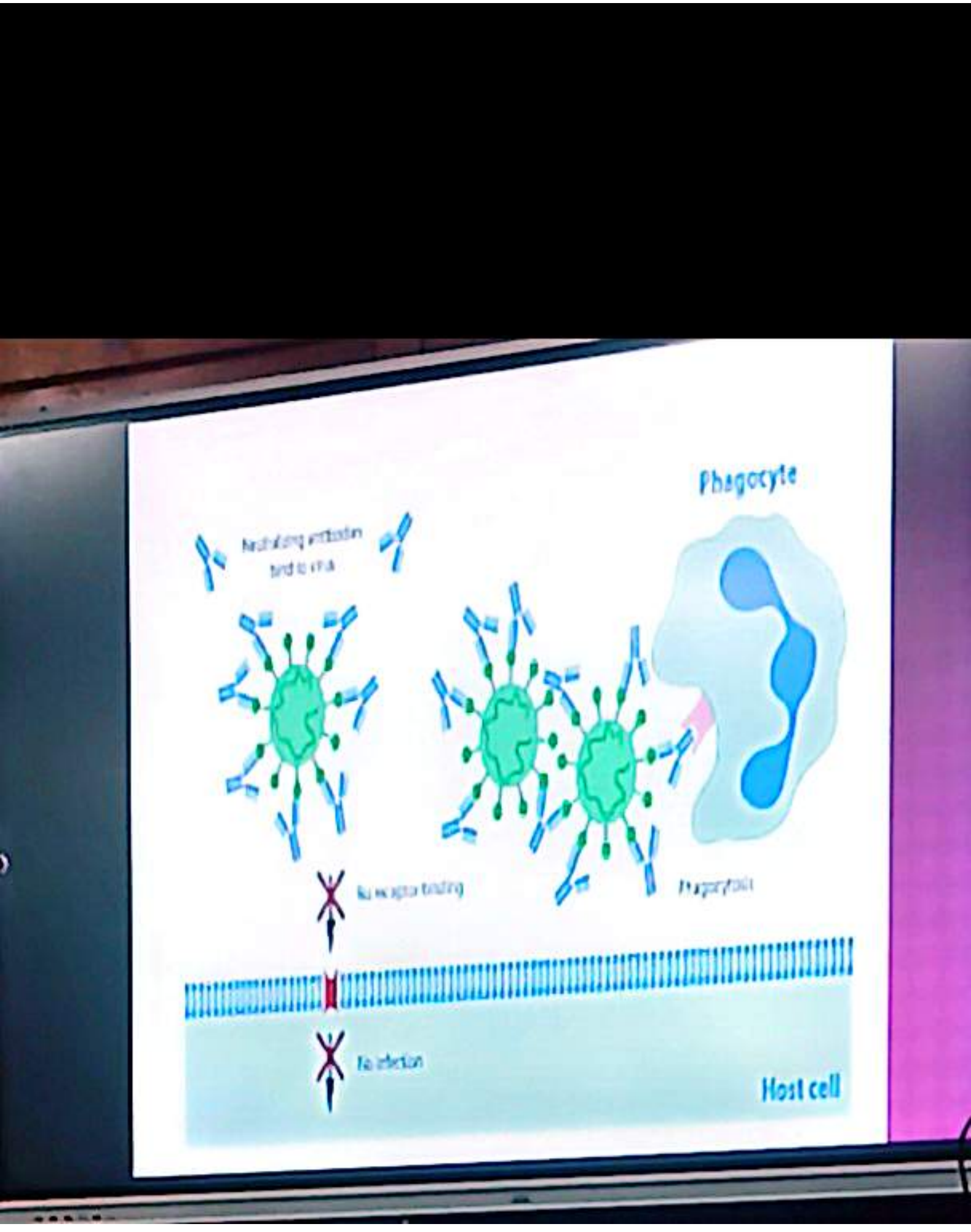
soluble molecules are cross-linked into large insoluble complexes that are insoluble.

5. Complement activation:

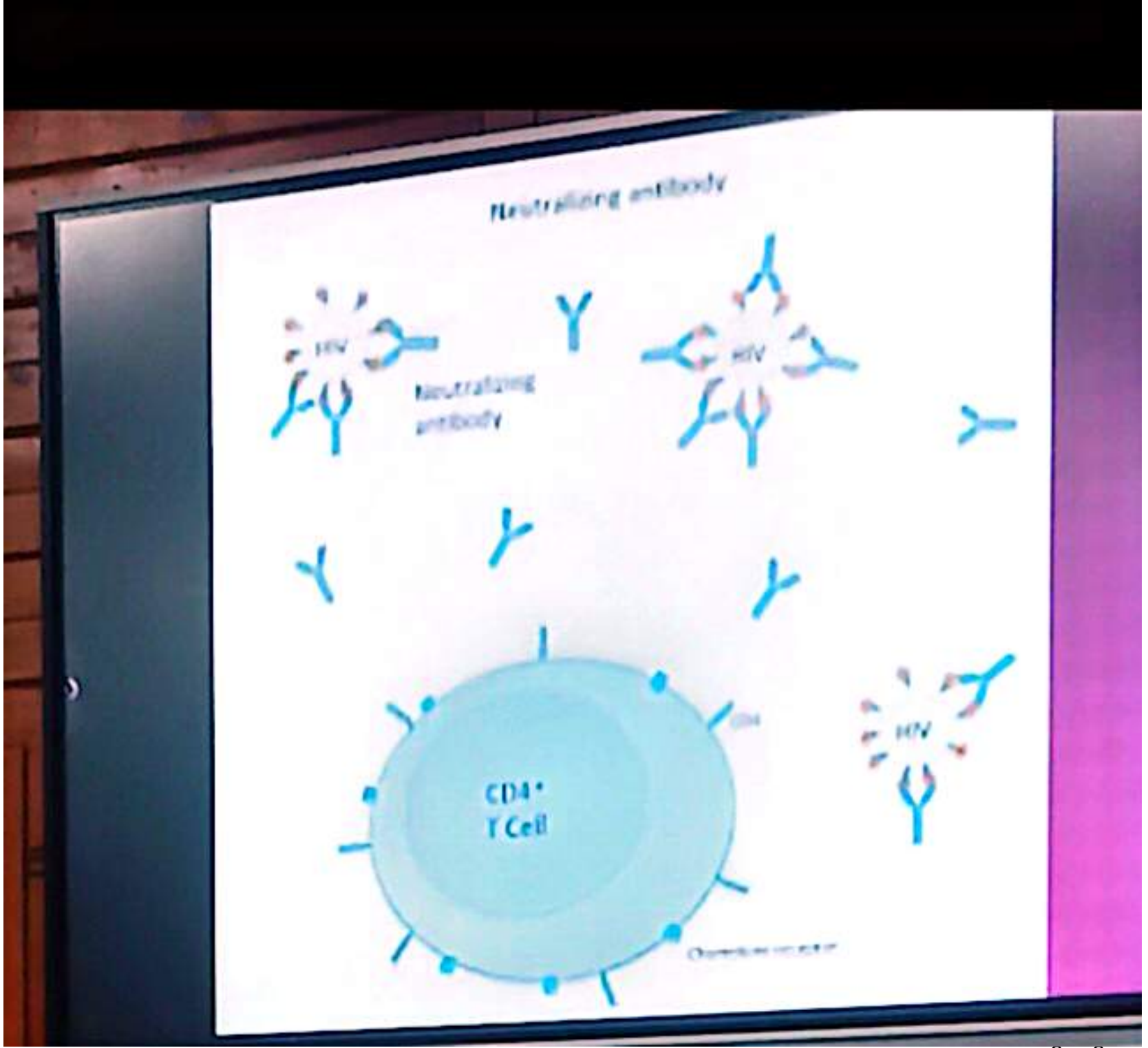
is the main mechanism used against cellular antigens.

6. Trigger mast cell degranulation.

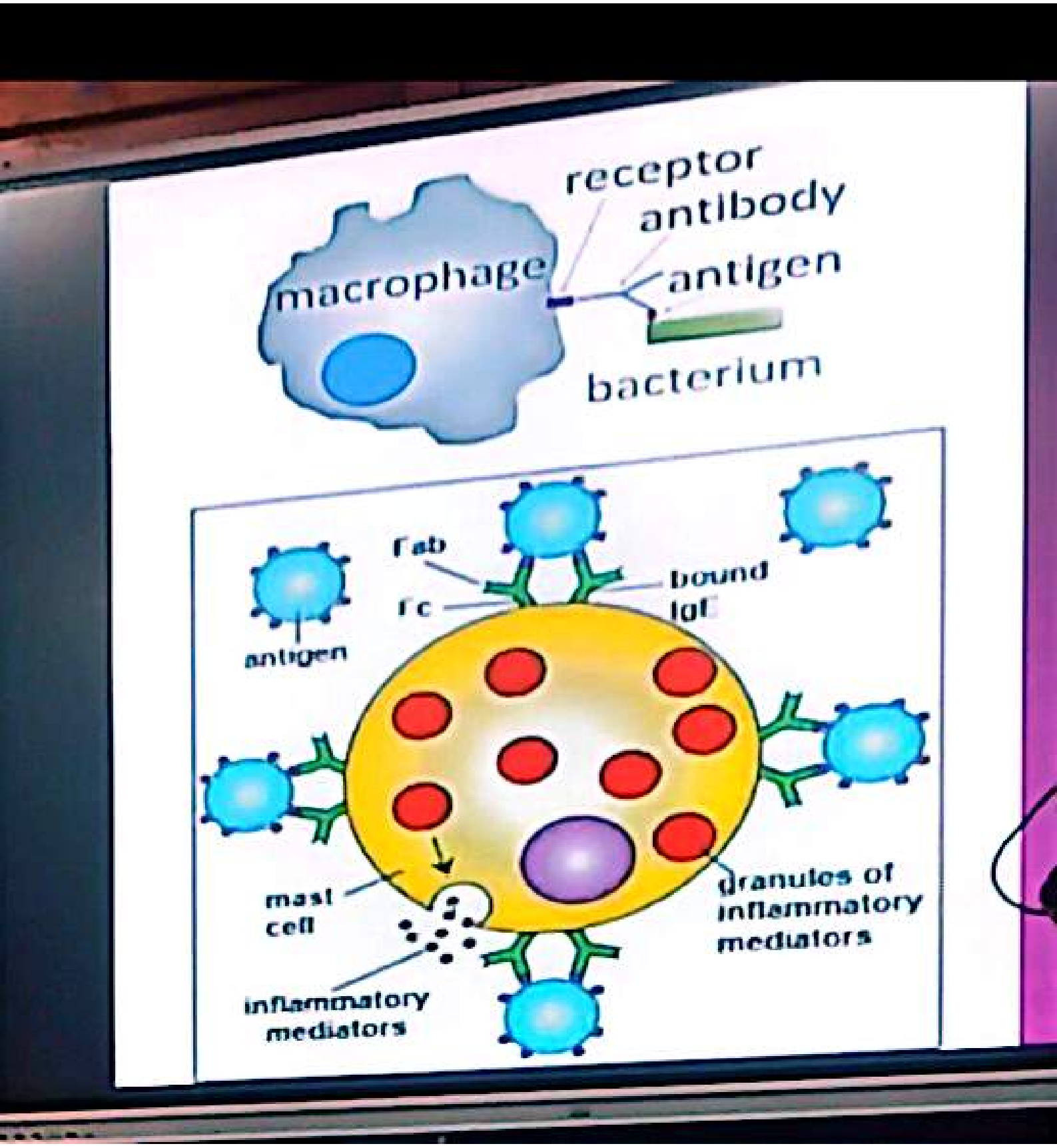
تم المسح الضوئي بـ CamScanner



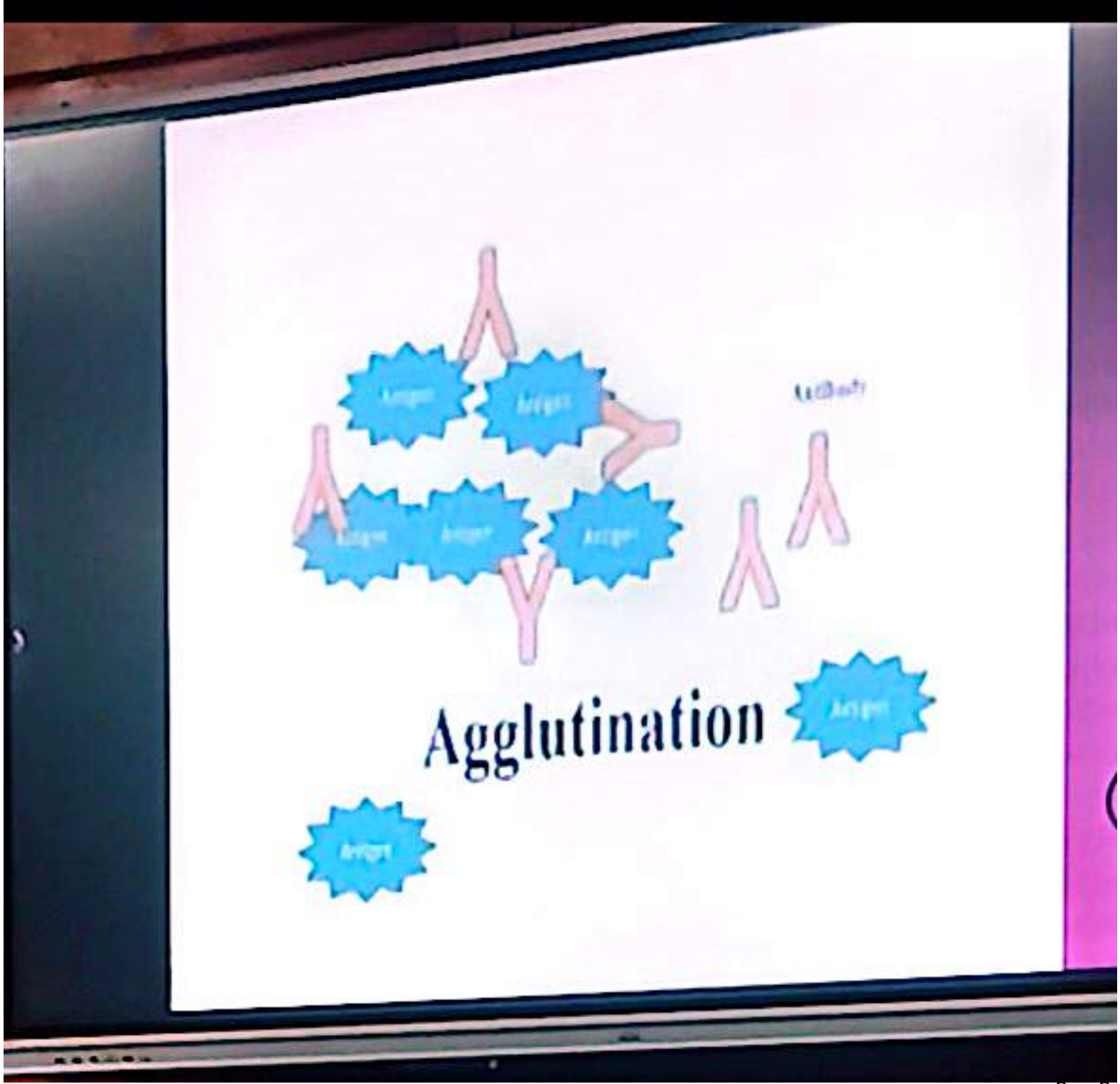
تم المسح الضوئي بـ CamScanner



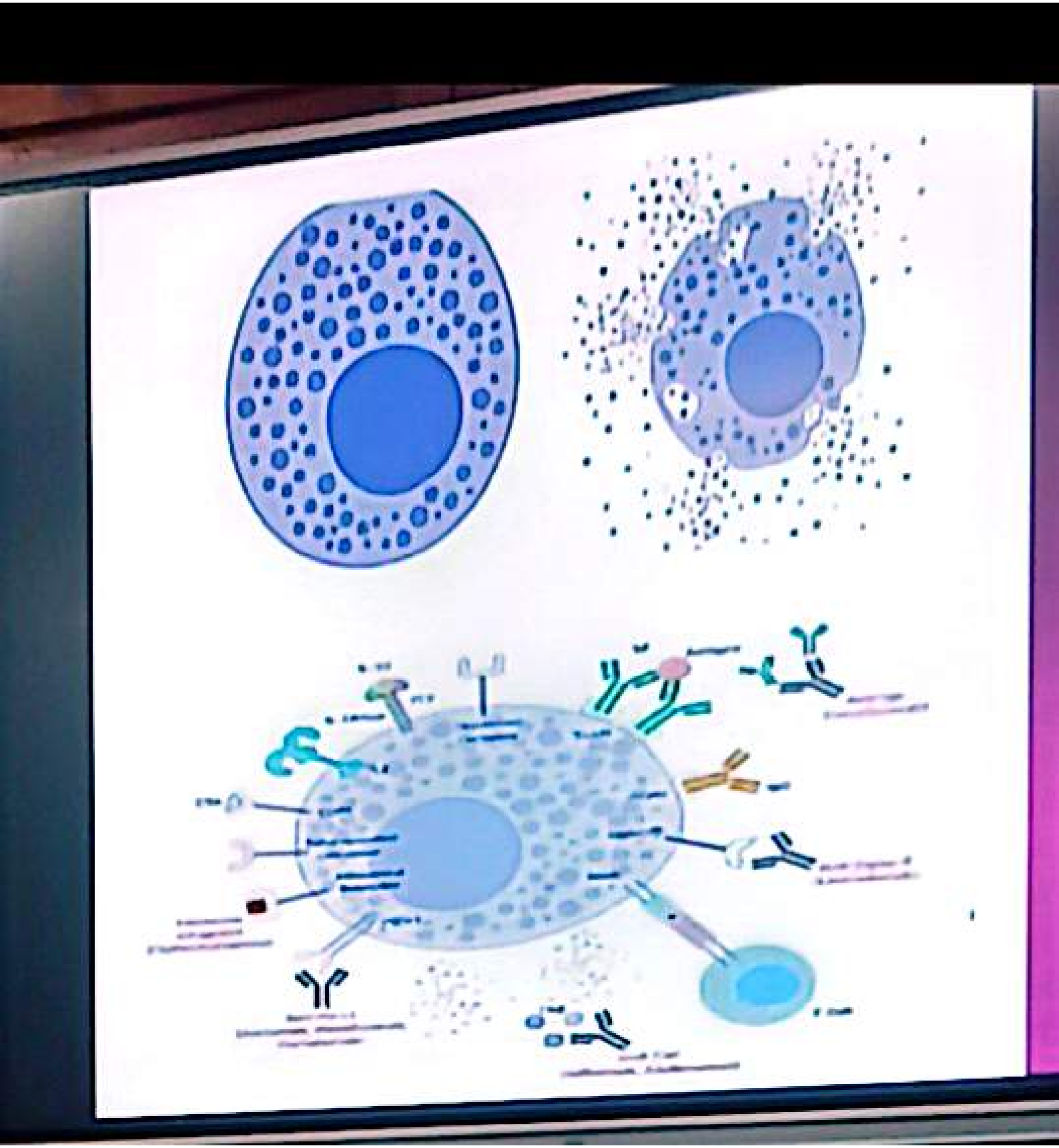
تم المسح الضوئي بـ CamScanner



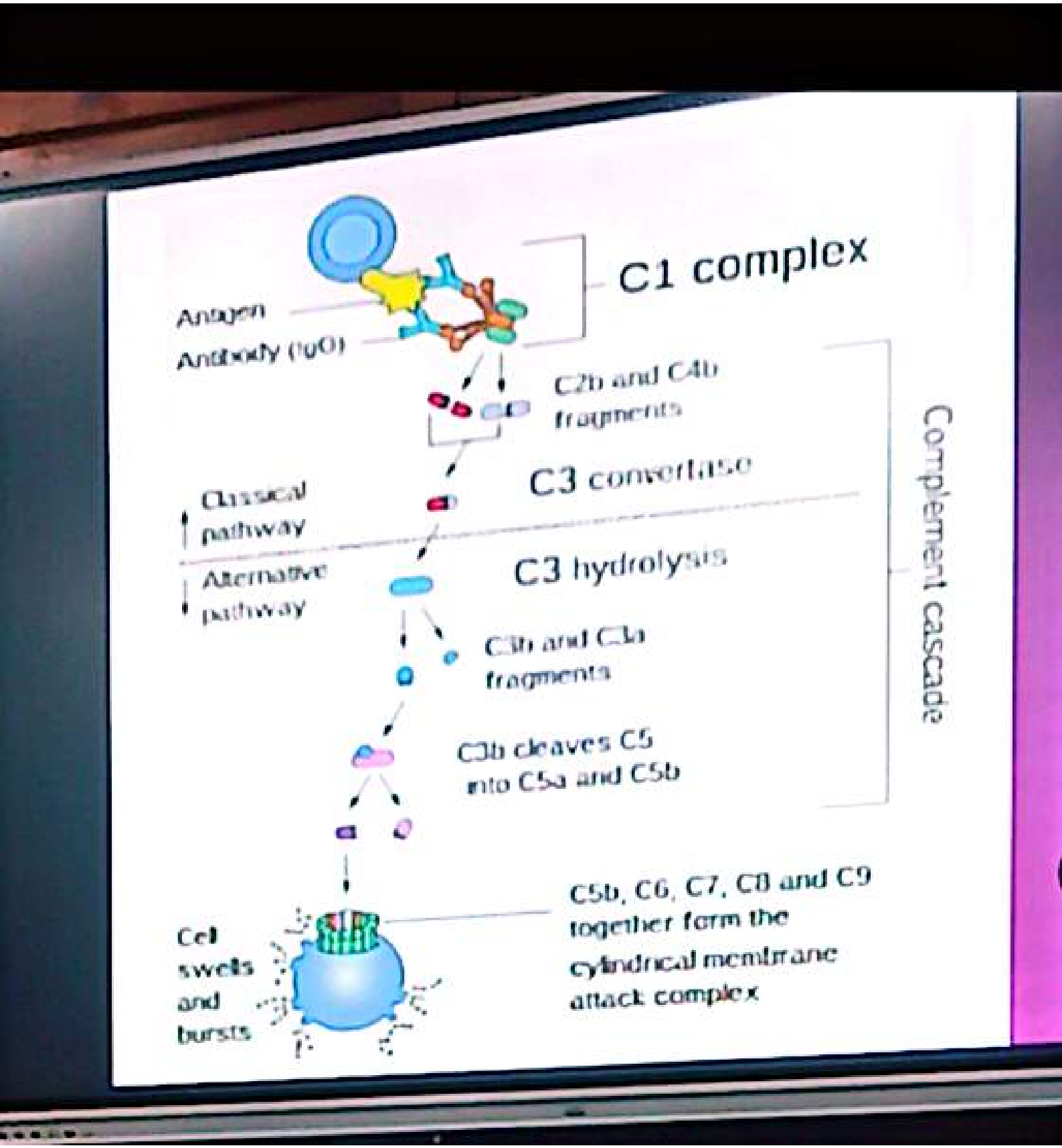
تم المسح الضوئي بـ CamScanner



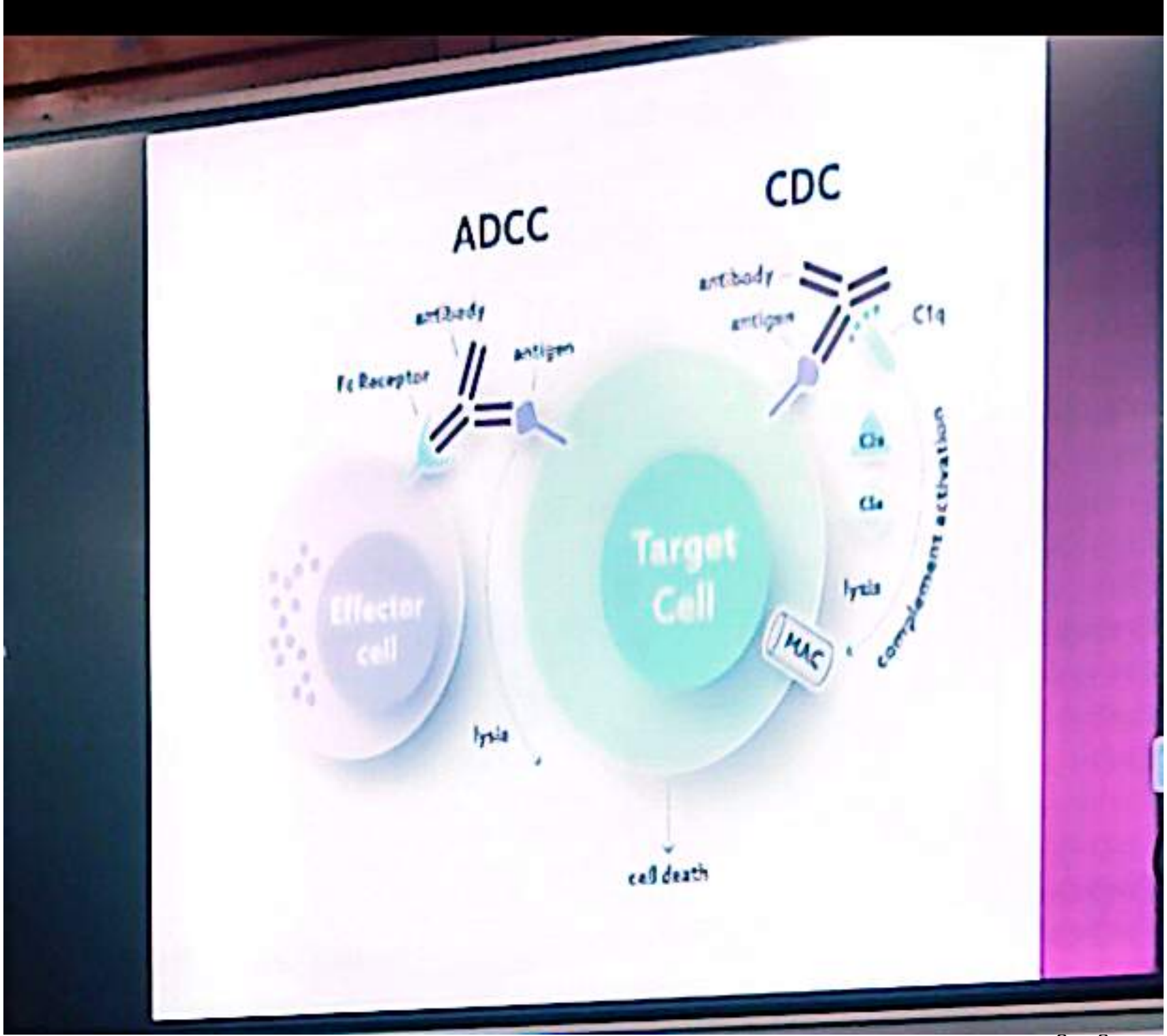
تم المسح الضوئي بـ CamScanner



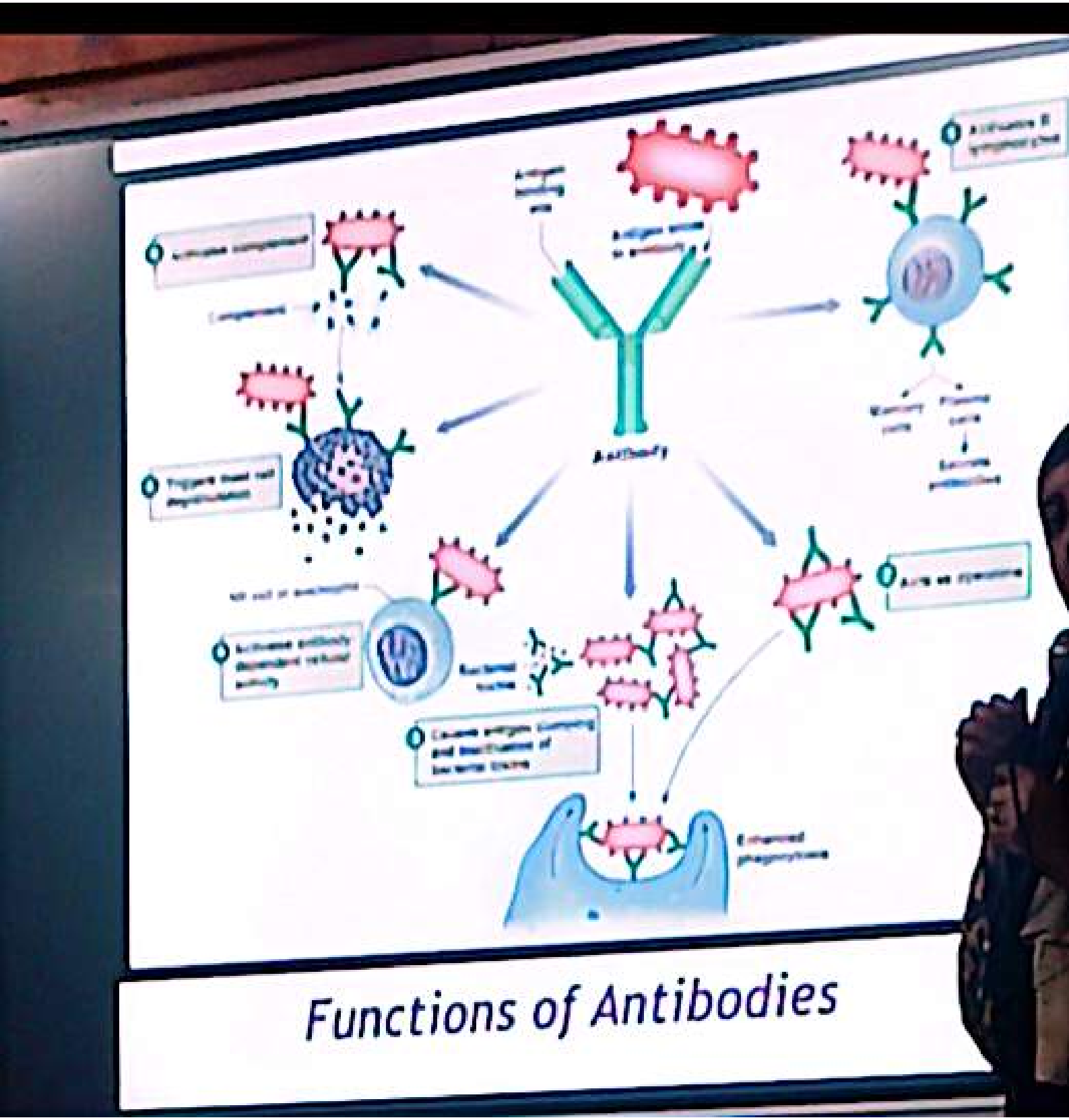
تم المسح الضوئي بـ CamScanner



تم المسح الضوئي بـ CamScanner



تم المسح الضوئي بـ CamScanner



تم المسح الضوئي بـ CamScanner

ANTIGEN-ANTIBODY INTERACTIONS

The Nature of Ag-Ab interactions are characterized by:

A. Lock and Key Concept:

The antigenic determinant of Ag nestles in a cleft formed by the combining site in Ab. Thus, Ag acts as a key which fits into a lock (*i.e.* specific Ab).

B. Non-covalent Binding:

The bonds that hold the Ag in the antibody combining site are all non-covalent in nature, including: hydrogen bonds, electrostatic bonds,

تم المسح الضوئي بـ CamScanner

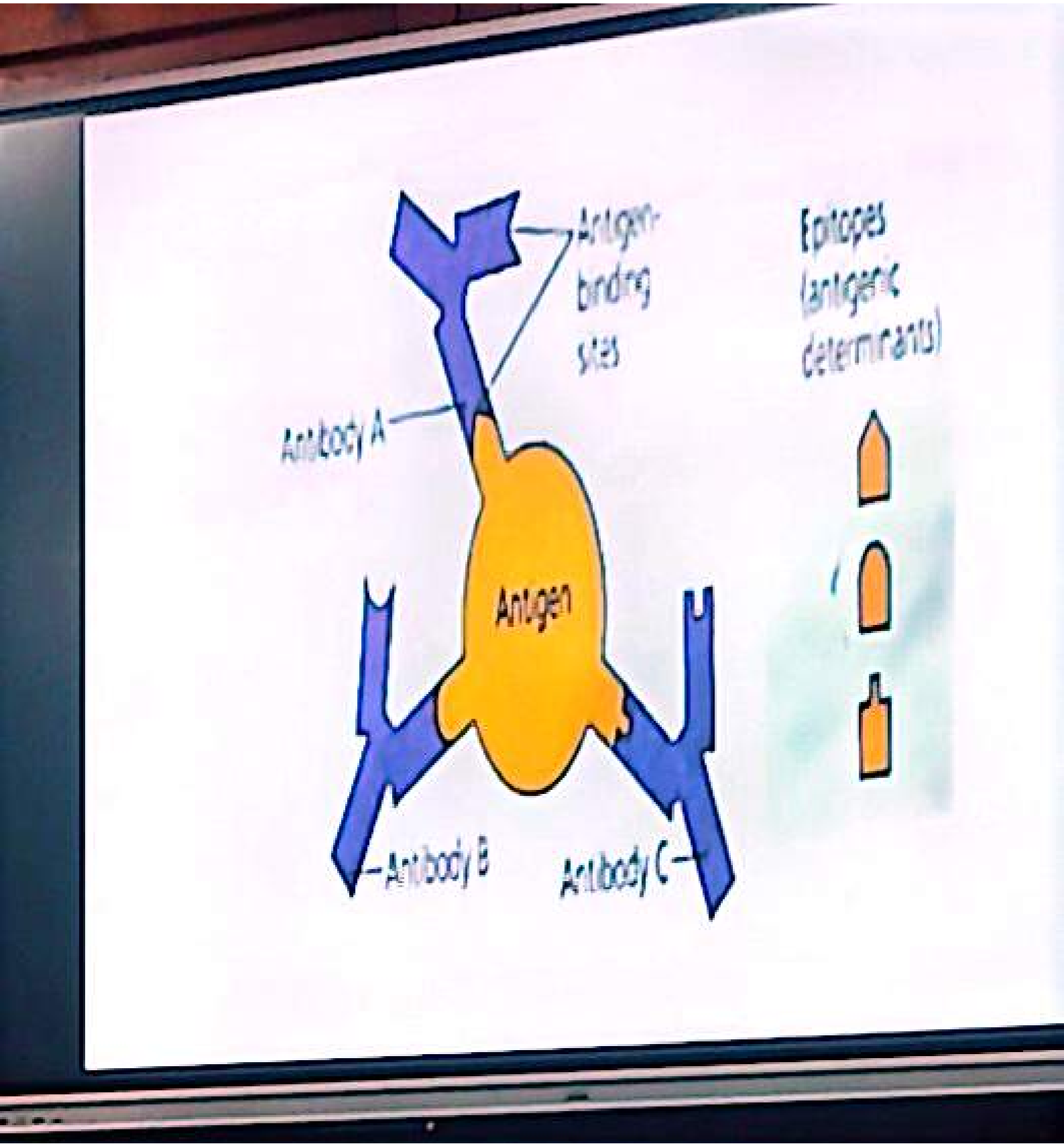
C. Reversability:

Since Ag-Ab reactions occur via non-covalent bonds, binding will be reversible.

D. Specificity:

the ability of an individual antibody combining site to react with only one antigenic determinant or the ability of a population of antibody molecules to react with only one antigen. The specificity of an antibody for a corresponding antigen is determined by the size and shape of both the antigenic determinant and Fab

تم المسح الضوئي بـ CamScanner



تم المسح الضوئي بـ CamScanner

- The Fc portion of Ig E usually:
- a. Binds to an Fc receptor on mast cell
- b. highly variable
- c. Part of antigen recognition and binding.
- d. Mediates complement fixation)



تم المسح الضوئي بـ CamScanner

Summary

- Immunoglobin are five classes
- Each class of immunoglobulin has different function
- Ag /Ab binding is reversible

تم المسح الضوئي بـ CamScanner